EPIDEMIOLOGICAL MACHINE LEARNING • POPULATION SCREENING TRIAGE

When Training Labels Change Who Gets Screened

Temporal evaluation of diagnosis-based versus biomarker-augmented labels in NHANES under finite screening capacity

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01 THE CLINICAL PROBLEM

Screening triage algorithms often train on self-reported prior diagnoses or electronic health record proxies. However, clinical diagnoses require regular clinical encounters, misclassifying underserved individuals facing access barriers as healthy baselines.

02 STUDY DESIGN & TEMPORAL PROTOCOL

We evaluate whether augmenting self-reported diagnosis with objective laboratory biomonitoring alters triage priority, holding non-invasive predictors and screening capacity constant.

03 METHODS & IMPLEMENTATION NOTE

Primary models are **L2-regularized logistic regressions** tuned via 5-fold cross-validation on median-imputed, standard-scaled anthropometrics (age, sex, BMI, waist circumference). Poverty Income Ratio (PIR) and insurance coverage were strictly excluded from model inputs and reserved solely for equity audits. Data pipeline ingested 10 XPT files across two survey eras using an embedded DuckDB lakehouse; Pandera contracts validated biological ranges (HbA1c: 2.0% to 25.0%) and confirmed 0 pregnant participants.

KEY TAKEAWAY **Changing training labels altered screening priorities and identified two additional potential cases at the same screening capacity.**

04 PAIRED-SELECTION QUEUE ANALYSIS (20% CLINIC CAPACITY)

Fixed Capacity K = 939 slots / 4,698 adults

Holding capacity fixed at 20% (939 slots), both models prioritize 878 shared individuals. Priorities diverge for 122 individuals at the decision threshold:

Shared Selection (Both Models)
878 Individuals
52 Potential Cases
Positivity Rate: 5.92% (52 / 878)
High-risk concordance across both label strategies

Model A Exclusive (Diagnosis Only)
61 people • 1 case (1.64% yield)

Model B Exclusive (Biomarker Augmented)
61 people • 3 cases (4.92% yield)

Overall 20% Screening Yield (939 Slots):
Model A: 53 cases (5.64% PPV) → Model B: 55 cases (5.86% PPV) • **Net Gain: +2 Cases (+1.72 pp sensitivity)**

Note on precision: The 3.0× yield difference (4.92% vs 1.64%) applies strictly to the 61-person boundary swap, not to overall model precision (5.86% vs 5.64% unweighted). Overall case detection increased from 53 to 55 (+2 potential cases).

BENCHMARK PERFORMANCE & COMPLEX SURVEY VARIANCE (20% CAPACITY)

Screening Model	Slots (K)	Unweighted Cases	Unweighted Sens	Survey-Weighted Sens (95% CI)	Survey-Weighted PPV (95% CI)
Model A (Diagnosis Only)	939	53 / 116	45.69%	40.08% [28.9%, 51.2%]	5.79% [3.6%, 8.0%]
Model B (Biomarker Augmented)	939	55 / 116	47.41%	42.47% [28.4%, 56.5%]	6.02% [3.4%, 8.6%]
Net Change (Δ)	0	**+2 Cases**	**+1.72 pp**	**+2.38 pp** (overlapping CIs)	**+0.22 pp** (overlapping CIs)

Survey inference: Weighted estimates use Taylor Series Linearization (WTPH2YR weights, df = 15). The observed point-estimate difference is small (+2 cases, +2.38 pp weighted sensitivity), and formal paired confidence intervals accounting for complex survey clustering have not been evaluated.

05 HEALTH EQUITY SUBGROUP AUDIT

Evaluating detection across insurance coverage at 20% budget:

Subgroup	Total Cases	Model A Detection	Model B Detection
Insured Adults	102 cases	51 / 102 (50.0%)	52 / 102 (51.0%)
Uninsured Adults	**13 cases**	**2 / 13 (15.4%)**	**3 / 13 (23.1%)**

Small denominator note: The uninsured cohort contains only **13 latent cases**; detection changed by +1 case (2/13 to 3/13). While directionally consistent with hypothesis, small cell counts preclude claims of disparity elimination.

06 MULTI-ARCHITECTURE ROBUSTNESS

Re-evaluated triage using alternative decision tree algorithms:

- **Random Forest:** Model A detects 60 cases; Model B detects 61 cases (+1 case, 77 swaps).
- **HistGradientBoosting:** Model A detects 59 cases; Model B detects 59 cases (0 net gain, 132 swaps).

Confirms priority shifts occur across architectures, while overall yield gains remain modest (+0 to +2 cases).

07 LIMITATIONS & SCOPE

- **Surveillance vs Diagnosis:** Single venous HbA1c ≥ 6.5% represents an elevated screening criterion, not an autonomous clinical diagnosis.
- **Inference Bounds:** Separate survey-weighted confidence intervals overlap; paired variance estimation is required for confirmatory testing.
- **Operational Scope:** Assumes 100% screening adherence among prioritized adults and uniform laboratory costs.

References: ADA Standards of Care (2024) • CDC NCHS NHANES Continuous Biomonitoring (2017–2023) • Obermeyer et al., *Science* (2019) • Pre-specified observational protocol

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